1. **Question bank**
2. Define Artificial Intelligence and discuss its scope in real-world applications.
3. Explain how AI can be used to solve real-world problems. Provide an example.
4. What are the key challenges in AI problem-solving?
5. Briefly describe the major milestones in the history of AI.
6. Discuss the role of the Turing Test in AI development.
7. How did the emergence of deep learning influence the evolution of AI?
8. Differentiate between Narrow AI and General AI with suitable examples.
9. Discuss the ethical implications of developing General AI.
10. What are the key components of a well-defined AI problem?
11. Explain the difference between deterministic and stochastic problem-solving approaches.
12. How does constraint satisfaction help in AI problem-solving?
13. Compare uninformed search and informed search strategies with examples.
14. Describe the Breadth-First Search (BFS) algorithm and its advantages and disadvantages.
15. Explain the A* search algorithm with an example.
16. What is a heuristic function? How does it improve search performance?
17. Explain the difference between local search and global search in AI.
18. How does the Min-Conflicts heuristic help in solving constraint satisfaction problems?
19. Describe how Genetic Algorithms are used for heuristic search in AI.
20. Solve a simple constraint satisfaction problem using backtracking.

Module 2 Knowledge Representation and Reasoning (KRR)

21. Define Knowledge Representation (KR) and explain its importance in AI.
22. What are the different characteristics of a good knowledge representation system?
23. Explain the various approaches used in Knowledge Representation.
24. Differentiate between declarative and procedural knowledge with examples.
25. Describe the four main types of knowledge representation in AI.
26. Compare **semantic networks** and **frames** in knowledge representation.
27. What is a production rule? How is it used in knowledge representation?
28. Define Propositional Logic and explain its syntax and semantics with an example.
29. Convert the following English sentence into propositional logic: *"If it rains, the ground will be wet."*
30. What are the limitations of propositional logic? How does FOL overcome them?
31. Explain the key components of **First-Order Logic (FOL)** with examples.
32. Convert the following statement into **First-Order Logic**: *"All humans are mortal, and Socrates is a human."*
33. Discuss the **unification algorithm** in First-Order Logic.
34. What is a **semantic network**? Explain with an example.

35. How are **inheritance properties** handled in semantic networks?
36. Explain the concept of **frames** in knowledge representation with an example.
37. How do **frames** help in storing structured knowledge?
38. Compare **semantic networks** and **First-Order Logic** in terms of knowledge representation.
39. What are the limitations of **semantic networks** in AI?